

SCQ (Single Correct Type) :

1. 2 litres of an acidified solution of KMnO_4 , containing 1.58 g of KMnO_4 per litre, is decolorized by passing sufficient amount of SO_2 gas. If whole of the sulphur from x g of FeS_2 is converted into SO_2 to be used in above reaction, calculate the value of x :
- (A) $x = 1.5$ (B) $x = 3$ (C) $x = 4.5$ (D) $x = 6$
2. In 1L saturated solution of CuCl , 0.1 mol AgCl is added. K_{sp} of $\text{AgCl} = 1.6 \times 10^{-10}$. If the resultant concentration of Ag^+ in the solution is 1.6×10^{-7} , then K_{sp} for CuCl is :
- (A) 1.6×10^{-3} (B) 10^{-3} (C) 10^{-6} (D) 1.6×10^{-6}

Paragraph for Question Nos. 3 to 4

Zelina, a student of class XI is working in the chemistry lab of her school. She is provided with 4 containers of large capacity by the lab assistant.

Container 1 contains 2L of '2.8 V' H_2O_2 .

Container 2 contains 2L of '16.8 V' H_2O_2 .

Container 3 contains sufficient amount of water.

Container 4 is empty.

She has been asked by her teacher to prepare H_2O_2 solution using the components of container 1, 2 or 3 (partially or completely) and store it in container 4.

Now answer the following two questions :

3. The volume of water required by Zelina to prepare maximum volume of 2.55% (w/v) H_2O_2 solution is :
- (A) 0.33 L (B) 1 L (C) 0.67 L (D) 2 L
4. Zelina prepared a 12.6 V H_2O_2 solution and mixed it with excess of KI solution and titrated the liberated I_2 with $\frac{9}{7}$ M hypo solution. Find the maximum volume (in L) of hypo solution that could have been consumed in above process :
- (A) 5.44 L (B) 5 L (C) 4.67 L (D) 2.5 L

Paragraph for Question Nos. 5 to 6**Arrhenius theory of reaction rate :**

It gives insight in to the energetics and mechanistic aspects of reactions.

It is based upon kinetic theory of gases.

Arrhenius proposed a theory of reaction rate which states as follows :

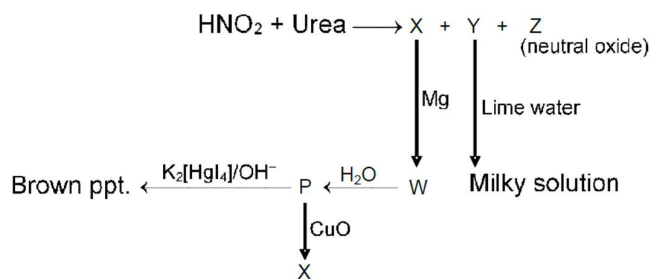
- ☐ A chemical reaction takes place due to the collision among reactant molecules
- ☐ Every collision does not bring a chemical change. The collision that actually produce the products are effective collision.

5. Select the correct statement(s) :
- (I) Arrhenius equation is not applicable for second order reaction.
 (II) Arrhenius equation is not applicable for radioactive decay
 (III) In Arrhenius equation $e^{-E_a/RT}$, represents the fraction of molecules undergoing collision.
 (IV) In Arrhenius equation $e^{-E_a/RT}$, represents the fraction of molecules which have sufficient energy for reaction.
- (A) I, II, III (B) I, III, IV (C) II, IV (D) II, III, IV
6. The rate of reaction in presence of catalyst at temp 27°C may be made equal to the rate of reaction in absence of catalyst but for this, we have to raise the temperature of reaction upto 47°C and activation energy in absence of catalyst was found 9.6 KJ/mole. What should be activation energy in presence of catalyst.
- (A) 10.2 KJ/mole (B) 0.6 KJ/mole (C) 9 KJ/mole (D) None of these

MCQ (Multiple Correct Type) :

7. Which of following statements are correct :
- (A) Under same temperature and pressure conditions, average speed of O_2 gas molecules will be greater than that of CO_2 molecules.
 (B) All molecules in a gas are moving with the same speed.
 (C) If an open container is heated from 300 K to 400 K, the fraction of air which goes out with respect to originally present is $\frac{1}{4}$.
 (D) If compressibility factor of a gas at STP is less than unity then its molar volume is less than 22.4 L at STP.
8. Which of the following statements are true about the alkali metals ?
- (A) All alkali-metal salts impart a characteristic colour to the Bunsen flame.
 (B) Among LiOH, CsOH, KOH, and RbOH, CsOH has the highest solubility in water.
 (C) Among the alkali metals, cesium is the most reactive.
 (D) The reducing character of the alkali metal hydrides follow the order : $LiH > NaH > KH > RbH > CsH$.
9. When N_2 gas is adsorbed on the surface of metallic iron, electron transfer occurs from the metal to N_2 . The true statement(s) regarding this adsorption is/are :
- (A) N_2 is chemisorbed.
 (B) N–N bond length is increased.
 (C) Occupancy of antibonding molecular orbitals in N_2 is increased.
 (D) Entropy is decreased.
10. A solution of an unknown sample on treatment with dilute hydrochloric acid gave a white precipitate. The white precipitate formed was added to hot water and filtered. The white precipitate remaining on the filter paper was readily soluble in ammonia solution. A few drops of potassium iodide solution were added to the hot water filtrate and a bright yellow precipitate was produced. Which ions could have been present in the unknown sample?
- (A) $Ag^+_{(aq)}$ (B) $Hg^{2+}_{(aq)}$ (C) $Pb^{2+}_{(aq)}$ (D) $Ba^{2+}_{(aq)}$

11. Which of the following compound(s) have two pie bond in it's structure ?



- (A) X (B) Y (C) W (D) P
12. 50ml of 20.8% w/v BaCl_2 (aq) and 100 ml of 9.8% w/v H_2SO_4 (aq) solution are mixed. Then in final solution :
- (A) $[\text{Cl}^-] = 0.66\text{M}$ (B) $[\text{H}^+] = 1.33\text{M}$ (C) $[\text{Ba}^{2+}] \approx 0\text{M}$ (D) $[\text{SO}_4^{2-}] = 0.33\text{M}$
13. The pH of a 0.01 M acid HX is 2 and pH of 0.01 M salt ACl is also 2. What conclusions cannot be drawn from this information?
- (A) HX is a weak acid and AOH is strong base.
 (B) HX is a strong acid and AOH is strong base.
 (C) HX is a strong acid and AOH is very weak base.
 (D) HX is a strong acid and A^+ undergoes partial hydrolysis.
14. Consider the following statements :
- (I) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in molar concentration of each gaseous substance involved.
 (II) An increase in pressure (caused by decrease in volume) at equilibrium results in increase in no. of moles of each gaseous substance involved.
 (III) For the reaction $\text{A(s)} \rightleftharpoons \text{B(g)} + \text{C(g)}$ starting with A only at constant temperature, total pressure at equilibrium (of B(g) & C(g)) is not affected by changing the volume of container.
 (IV) $\text{H}_2\text{O(l)} \rightleftharpoons \text{H}_2\text{O(g)}$; on temperature increase, K_p of this reaction is increased.
- Tick the correct statement(s)
- (A) (I) (B) (II) (C) (III) (D) (IV)
15. Which of the following pairs of coordination compound show same type of isomerism?
- (A) $[\text{Co}(\text{NH}_3)_5\text{Br}] \text{SO}_4$ and $[\text{Ni}(\text{dmg})_2]$ (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Co}(\text{en})_2\text{BrCl}]\text{Cl}$
 (C) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)]$ (D) $[\text{Ni}(\text{en})_3]^{2+}$ and $[\text{Co}(\text{ox})_3]^{3-}$
16. In which of following silicate(s), the number of corners shared per tetrahedron is ≥ 2 ?
- (A) Four membered cyclic silicate (B) Pyrosilicate
 (C) Chain silicate (D) 2-D silicate
17. Which of the following is/are correct?
- (A) $\square$ Vander Waal's radius of iodine is more $\square$ than its covalent $\square$ radius
 (B) All isoelectronic ions belong to the same period of periodic table
 (C) First ionisation energy of Be is more than that of C
 (D) Electron affinity of N as well as noble gases is negative.

Match the column

18. Match the following :

List-I

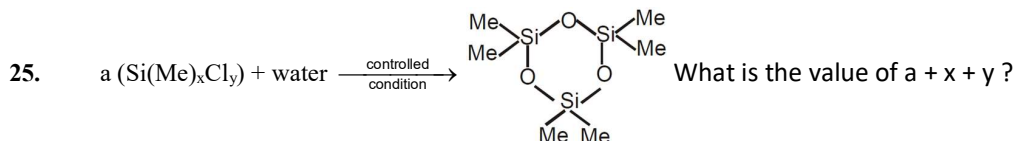
- (P) For a Vander Waal's gas with $b \approx 0$, $a \neq 0$
 (Q) For a real gas at its Boyle's temperature in high pressure region
 (R) For a Vander waal gas with very large molar volume
 (S) For a Vander waal gas at the critical point

List-II

- (1) $Z = 0.375$
 (2) $Z < 1$
 (3) $Z > 1$
 (4) $Z = 1$

(Numerical Value type) :

19. (i) Number of P-P bonds in P_4S_3 is (A).
 (ii) Amongs the following number of planar species is (B).
 XeF_2 , ClF_3 , H_2O , $[XeF_5]^-$, I_3^- , BCl_3 , XeF_4 , SF_4 , PCl_5 , SF_6 , IF_7 .
 Determine the value of (A + B).
20. A metal 'S' is added in excess to 500 mL of 1 molar solution of another metal ion B^{n+} at $25^\circ C$ until the equilibrium, $S(s) + B^{n+}(aq) \rightleftharpoons S^{n+}(aq) + B(s)$ is reached. If $E_{S^{n+}/S}^\circ$ and $E_{B^{n+}/B}^\circ$ are -0.75 V and -0.24 V respectively. Given at equilibrium, the concentration of B^{n+} is 5×10^{-18} M. The value of n is:
 Given ($\log 5 = 0.70$ and $\log 2.5 = 0.40$) ($\frac{2.303RT}{F} = 0.06V$). Give your answer to the nearest single digit integer.
21. Given two separate reactions:
 (I) $A \longrightarrow \text{Product}$
 (II) $B \longrightarrow \text{Product}$,
 Reaction (I) follows first order kinetics while reaction (II) follows second order kinetics. Both the reactions have same half lives. When the initial concentration of (A) is 'n' times that of (B), then the rate of reaction (I) is 1.386 times than that of rate of reaction (II) at the start of reaction. The value of n is :
22. An organic compound contains C, H and O atoms. One molecule of the compound contains H-atoms equal to 66.67 % of total atoms and mass ratio of C to O is 3 : 2. If the molecular formula of the compound is $C_xH_yO_z$, what is the value of X + Y + Z. (Given vapour density of compound is 23 g/mol)
23. During the preparation of $H_2S_2O_8$ (peroxydisulphuric acid) using H_2SO_4 electrolytically, O_2 gas also releases at anode as by product. When 10.304 L of H_2 releases at cathode and 2.24 L O_2 at anode at STP, the weight of $H_2S_2O_8$ produced in gram is : (Round it off to nearest whole number)
24. How many of the following compound can be optically active?
 (a) $[Co(en)_2ClBr]$ (b) $[Co(en)(NH_3)_2Cl_2]^+$ (c) $[Cr(NH_3)_4Cl_2]^+$ (d) $[Co(NH_3)_3Cl_3]$
 (e) $[Co(en)_3]^{3+}$ (f) $[Ni(dmg)_2]$ (g) $[Pb(EDTA)]^{2-}$ (h) $[Cr(ox)_2(dmg)]^{2-}$
 (i) $[Cr(dmg)_3]$



Answer Key

1.	(B)	2.	(C)	3.	(C)	4.	(B)	5.	(C)	6.	(C)
7.	(ACD)	8.	(ABC)	9.	(ABD)	10.	(AC)	11.	(AB)	12.	(ABC)
13.	(ABD)	14.	(CD)	15.	(BD)	16.	(ACD)	17.	(AD)		
18.	(A) – Q, (B) – R, (C) – S, (D) – P					19.	10.00	20.	02.00	21.	02.00
22.	9	23.	50	24.	6 (a, b, e, g, h, i)	25.	7				

